

# Patrick Kramer

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## EDUCATION

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| <b>Carnegie Mellon University</b><br>PhD in Statistics   <u>GPA: 3.99/4.00</u><br>Advisor: Edward H. Kennedy                                                                                                    | Pittsburgh, PA<br>Expected: May 2029 |
| <b>Leipzig University</b><br>MSc equivalent (Diplom) in Mathematics with distinction   <u>Class Rank: 1 of 65</u><br>Thesis: Regression Discontinuity Designs with Covariates (Advisor: Alexander Fuchs-Kreiss) | Leipzig, Germany<br>July 2024        |
| <b>Leipzig University</b><br>BSc equivalent (Vordiplom) in Mathematics   <u>Class Rank: 1 of 65</u>                                                                                                             | Leipzig, Germany<br>May 2022         |
| <b>Leipzig University</b><br>BSc in Computer Science   <u>Class Rank: 1 of 252</u><br>Thesis: A Generalization of the Iteration Theorem for Recognizable Formal Power Series (Advisor: Andreas Maletti)         | Leipzig, Germany<br>September 2019   |

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## RESEARCH INTERESTS

I am broadly interested in Machine Learning, Causal Inference, and High-Dimensional Statistics, and particularly their intersection. Most recently, I have been working on causal inference with high-dimensional treatments.

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## RESEARCH EXPERIENCE

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| <b>Carnegie Mellon University</b><br>PhD Researcher (with Edward Kennedy)<br>Project: “Causal Inference with High-Dimensional Treatments” <ul style="list-style-type: none"><li>Develop pseudo-outcome regression framework for efficient estimation of high-dimensional target parameters that allows the use of flexible nonparametric machine learning methods for nuisance parameter estimation</li><li>Derive double machine learning estimators to learn mean potential outcomes and incremental effects for high-dimensional treatments based on the proposed framework, and prove minimax optimality</li><li>Apply proposed methods to evaluate the impact of public schools in North Carolina on students’ End-of-Grade test results in mathematics and reading based on a dataset of the North Carolina Education Research Data Center</li></ul> | January 2025 - present<br>Pittsburgh, PA       |
| <b>Leipzig University</b><br>Graduate Student Researcher (with Alexander Fuchs-Kreiss)<br>Project: “Estimating ATEs in Regression Discontinuity Designs with Covariates under Minimal Assumptions” <ul style="list-style-type: none"><li>Proved asymptotic normality of the covariate-adjusted estimator of the average treatment effect in regression discontinuity designs to conduct a sensitivity analysis in the case of confounding</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                        | August 2022 - October 2023<br>Leipzig, Germany |

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## INVITED TALKS

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| <b>Joint Statistical Meetings (JSM) – Session for Winners of Student Best Paper Award</b><br>Causal Inference with High-Dimensional Treatments | Boston, MA<br>August 2026   |
| <b>INFORMS Annual Meeting</b><br>Causal Inference with High-Dimensional Treatments                                                             | Atlanta, GA<br>October 2025 |

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## PAPERS

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Causal Inference with High-Dimensional Treatments (2026) – <b>JSM Student Best Paper Award</b><br><u>Patrick Kramer</u> , Edward H. Kennedy, Isaac M. Opper | <a href="#">[arXiv]</a><br>preprint |
| Estimating ATEs in Regression Discontinuity Designs with Covariates under Minimal Assumptions (2023)<br><u>Patrick Kramer</u> and Alexander Fuchs-Kreiss    | <a href="#">[arXiv]</a><br>preprint |

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## WORK EXPERIENCE

### PricewaterhouseCoopers (PwC)

Associate Finance Transformation

Leipzig, Germany

February 2022 - June 2024

- Consulted on data migration for SAP S/4HANA Cloud implementation projects, such as for a major American aerospace company, a large German automobile company, and a media company, to successfully implement new cloud-based SAP systems
- Organized and led business data migration workshops for the clients to ensure a seamless Go-Live of new SAP S/4HANA Cloud systems for a total of three large corporations
- Created data migration concepts, test cases, and cut-over plans to ensure a complete and timely data transfer from the old to the new system and guarantee the reliability of the newly implemented SAP system
- Performed data reconciliation after the data migration using Python, R, and Excel to guarantee that the transferred data was complete and correct; documented data migration activities for subsequent audits

### Check24

Front End App Developer for Android

Leipzig, Germany

May 2020 - January 2022

- Developed the front end of a mobile app for Android using object-oriented programming with Kotlin and Java to compare flight and hotel prices from various tour operators
  - Handled API requests to the back end for requested price comparisons using Postman to accommodate user search requests
  - Programmed test scripts using Appium to automate user interface testing for Android and iOS; set up Bamboo script to automatically execute the tests before deployment
  - Collaborated with around 15 team members using the version control system Git to organize the source code
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## TEACHING EXPERIENCE

### Carnegie Mellon University

Teaching Assistant - Introduction to Probability Modeling

Pittsburgh, PA

Spring 2025

Teaching Assistant - Mathematical Statistics and Probability Theory

Fall 2024

### Leipzig University

Teaching Assistant - Calculus for Computer Scientists

Leipzig, Germany

Fall 2018

Teaching Assistant - Computer Hardware Engineering

Summer 2018

Teaching Assistant - Modeling and Programming

Summer 2018

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## HONORS AND AWARDS

JSM Student Best Paper Award (Statistical Learning and Data Science Section)

January 2026

Deutschlandstipendium | Awarded by German Federal Ministry of Education and Research

October 2021

Deutschlandstipendium | Awarded by German Federal Ministry of Education and Research

October 2019

Top Student Award | Awarded by Database Group Leipzig and Leipzig University

December 2018

Deutschlandstipendium | Awarded by German Federal Ministry of Education and Research

October 2018

Deutschlandstipendium | Awarded by German Federal Ministry of Education and Research

October 2017

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## ACADEMIC SERVICE

Co-Organizer - Causal Inference Reading Group, Carnegie Mellon University

August 2025 – present

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## SKILLS

- **Programming Languages:** R, Python, Kotlin, Java, C, C++, SQL, JavaScript, MATLAB
- **Technologies:** NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Git, Visual Studio Code, RStudio, LaTeX